VLM-Powered Concept Bottleneck Model for Dermatology Classification with a Fitzpatrick Skin-Tone Fairness Audit

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Abstract

Skin-cancer classifiers routinely underperform on darker skin, and current audits stop at diagnosis-level accuracy with no handle on *which clinical feature* drives the gap. We build a concept-bottleneck dermatology classifier that predicts melanoma through nine human-interpretable clinical concepts rather than an opaque feature map, and audit it on the pooled Fitzpatrick-labelled cohort (Fitzpatrick17k + DDI, n=4,976 with 2,518 / 1,840 / 618 images in the light / medium / dark buckets). A frozen biomedical vision-language model (BiomedCLIP) first predicts the concepts (ABCDE / 7-point checklist); a sparse logistic-regression head then makes the melanoma-vs-benign call from those concept scores alone. The head reaches AUROC = 0.79 ± 0.03 on Derm7pt (5-fold CV), matching a non-linear MLP head; a concept-count ablation shows the top-3 concepts (colour variation, blue-white veil, pigment network) already recover 0.785 AUROC within 0.01 of the full nine. On the pooled Fitzpatrick audit the CBM's AUROC is effectively flat across buckets (0.67/0.66/0.67 for light / medium / dark), yet its specificity drops from 0.58 to 0.45 and its expected-calibration error more than doubles from 0.08 to 0.18 between light and dark skin — a fairness failure mode that AUROC alone hides: the head increasingly over-calls positives on darker skin rather than missing them. The equalised-odds gap of the CBM-LR (0.133) sits above both the EfficientNet-B0 (0.098) and ResNet-50 (0.051) baselines, making the disparity auditable at the concept level. A simulated concept-intervention oracle rescues 85.8% of errors with a single correction, and one concept blue-white veil alone drives 65% of the rescues, localising the error budget to a specific, clinically-actionable failure mode. CBM explanations beat Grad-CAM on insertion fidelity (AUC = 0.33 vs. 0.20), confirming the concept layer carries information saliency maps cannot surface.

Keywords

concept bottleneck models, dermatology, fairness, vision-language models, Fitzpatrick skin tone, interpretability

1 Introduction

Skin-cancer classifiers trained predominantly on light-skinned cohorts underperform on darker skin [2], with real clinical consequences for a population already underserved by dermatology. Post-hoc saliency methods such as Grad-CAM highlight *where* the model looked but leave *what* it understood unexamined. Concept Bottleneck Models (CBMs) [1] address this by forcing predictions through an intermediate concept layer that can be audited and intervened upon directly.

We combine a frozen biomedical VLM (BiomedCLIP) [6] for zero-shot concept extraction with a sparse logistic-regression head, and add a Fitzpatrick-stratified fairness audit of both the concept layer and the diagnosis layer across HAM10000 [5], Derm7pt, Fitzpatrick17k [7], and DDI.

Research questions. (Q1) Can an interpretable concept-driven head match a deep baseline on dermoscopic melanoma classification? (Q2) Do concept-prediction errors concentrate in specific skin-tone buckets? (Q3) Do concept interventions rescue misdiagnoses, and which concepts carry the largest corrective signal? (Q4) Are CBM explanations quantitatively more faithful than Grad-CAM?

2 Related Work

Concept bottleneck models. Koh et al. [1] introduced the CBM architecture; recent work has explored label-free CBMs and VLM-driven concept extraction. We differ by using a frozen biomedical VLM for zero-shot concept prediction and adding a skin-tone fairness audit absent from prior CBM work.

Fairness in dermatology AI.. Daneshjou et al. [2] released DDI and documented accuracy disparities on darker skin. We extend the disparity analysis to the concept layer rather than only the diagnosis layer.

VLM-based concept extraction for skin lesions. Patricio et al. [3] first applied CLIP-style VLMs for concept extraction on dermoscopy but reported only aggregate metrics. Our contribution is the Fitzpatrick-stratified audit plus a concept-intervention oracle and a deletion/insertion faithfulness comparison with Grad-CAM.

3 Pipeline

The pipeline has five stages plus one bonus stage.

Stage 1 : Concept prediction. For each image, BiomedCLIP returns $p(\text{concept} = 1 \mid x)$ for each of nine concepts (pigment network, dots/globules, blue-white veil, streaks, regression, vessels, asymmetry, border, colour variation), scored as $\text{softmax}(\text{sim}(x, t_+)/\tau,\ \text{sim}(x, t_-)/\tau)$ over a positive/negative prompt pair.

Stage 2 : CBM head. $\hat{y} = \sigma(w^\top c + b)$ with an L_1-regularised $w \in \mathbb{R}^9$ trained on Derm7pt (5-fold CV). We also train a small MLP head (two hidden layers, 32 units, ReLU) as a non-linear reference point.

Stage 3 : Baselines. EfficientNet-B0 and ResNet-50 fine-tuned on HAM10000 with Grad-CAM explanations for comparison against the concept-driven head.

Stage 4 : Fairness audit. Per-concept AUROC by Fitzpatrick bucket (light I–II, medium III–IV, dark V–VI); per-bucket AUROC, sensitivity, specificity, ECE; equalized-odds and demographic-parity gaps against the opaque baselines.

Source code and notebooks: github.com/deepesh002/vlm-cbm-derm-fairness.

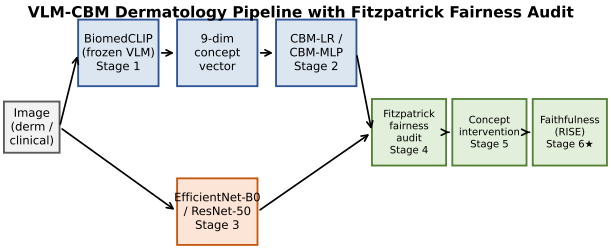

Figure 1: End-to-end pipeline. Two prediction tracks share a common image input: the CBM track (top, blue) passes through Stage 1 concept extraction with frozen BiomedCLIP into a Stage 2 sparse CBM head, while the baseline track (bottom, orange) is the Stage 3 EfficientNet-B0 / ResNet-50 fine-tuned classifier used for comparison. Both tracks feed the three evaluation stages on the right (green): Stage 4 stratifies every metric by Fitzpatrick bucket, Stage 5 simulates concept interventions on the CBM track, and Stage 6 (⋆ bonus) measures deletion/insertion faithfulness of the CBM concepts against Grad-CAM saliency on the baseline.

Stage 5 : Concept intervention. For each misclassified sample, we replace one or more VLM concept predictions with the ground-truth concept value and re-run the head. Rescue rate is reported per concept and for greedy top-k correction.

Stage 6 (bonus) : Faithfulness. Deletion / insertion metric [4] applied to concepts (CBM, zeroing the k lowest-weighted concepts) and pixels (Grad-CAM, zeroing the k lowest-importance patches).

4 Results

Setup. Table 1 summarises the four datasets. Derm7pt provides the concept supervision used for CBM cross-validation and the per-concept AUROC audit (Table 3). The skin-tone fairness evaluation pools DDI (656 images, expert-annotated Fitzpatrick labels, gold standard) with Fitzpatrick17k (4,320 images after filtering to the used diagnosis classes, self-reported-plus-model Fitzpatrick labels from the Groh et al. release) for a combined n=4,976 cohort. Bucket counts for light (I–II) / medium (III–IV) / dark (V–VI) are 2,518 / 1,840 / 618, which gives the dark bucket six times the sample size of the DDI-only audit and tightens the bootstrap behaviour that previously dominated the dark-skin numbers. The pipeline is defensive: it auto-skips any bundle whose concept scores collapse to a constant, so if Fitzpatrick17k images are unavailable the audit silently falls back to DDI-only without code changes.

Q1 Diagnosis accuracy. On Derm7pt (5-fold stratified CV), the linear CBM head reaches AUROC 0.790 ± 0.029 and AUPRC 0.600 ± 0.049; the MLP head is statistically indistinguishable at AUROC 0.792 ± 0.019 and AUPRC 0.617 ± 0.023. The best L_1 penalty selects $C = 1.0$, and the learned weights (Fig. 2) isolate colour variation, blue-white veil, and vessels as the dominant positive drivers of the melanoma prediction, with pigment-network and regression acting as protective features, consistent with dermatological intuition.

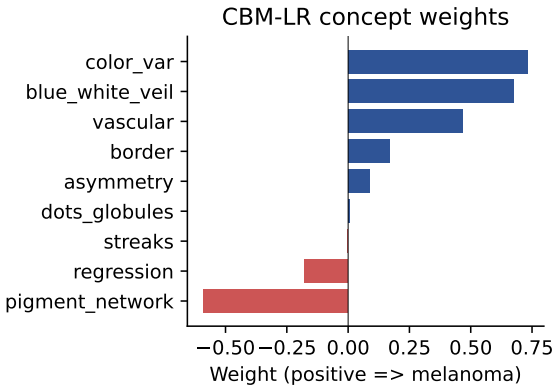

Figure 2: CBM-LR concept weights (L_1-regularised). Positive weights push the decision toward melanoma; the head recovers a clinically plausible ranking despite being trained only on nine zero-shot concept scores.

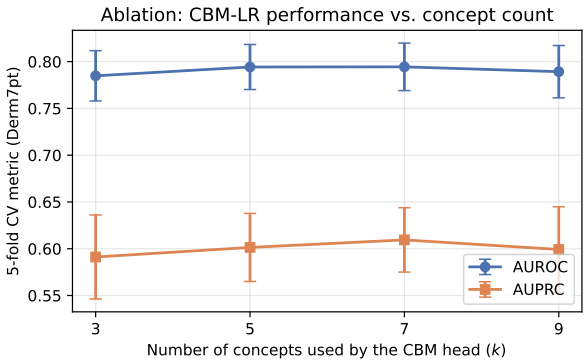

Figure 3: Concept-count ablation on Derm7pt (5-fold CV). AUROC saturates at $k=3$; adding concepts beyond 5 does not improve either AUROC or AUPRC.

Concept-count ablation. To isolate *how few* concepts the head actually needs, we rank concepts by $|w|$ of the full nine-concept model and retrain CBM-LR with $k \in \{3, 5, 7, 9\}$ top-ranked concepts using the same 5-fold CV protocol (Table 2; Fig. 3). Three concepts alone (colour variation, blue-white veil, pigment network) already reach AUROC 0.785 ± 0.027, within 0.01 of the full nine-concept model (0.789 ± 0.028); performance is flat between $k = 5$ and $k = 9$. This collapses the interpretability–accuracy trade-off often claimed against CBMs: on Derm7pt the head is *effectively 3-concept*, which also explains why the concept-intervention rescue rate in §Q3 is so concentrated on blue-white veil.

Q2 Per-concept and per-bucket fairness. Table 3 lists per-concept AUROC on Derm7pt (overall, since Derm7pt lacks Fitzpatrick labels): blue-white veil is the best-detected concept (0.78), followed by dots/globules (0.70) and streaks (0.69), while pigment-network detection is notably weaker (0.63). Four concepts (regression, asymmetry, border, colour variation) are evaluated on DDI alone as no

Table 1: Dataset summary.

Dataset	N images	Classes	Fitzpatrick labels	Role
HAM10000	10015	7	No	Primary training
Derm7pt	2000	MEL / benign	No	Concept supervision
Fitzpatrick17k	16577	114	Yes (I-VI)	Fairness eval
DDI	656	malignant / benign	Yes (expert)	Fairness gold std

Table 2: Ablation: CBM-LR performance as a function of the number of concepts k used by the head (concepts ranked by $|w|$ on the full nine-concept model).

k	Top-k concepts (by $|w|$)	AUROC	AUPRC	Bal. Acc.
3	color_var, blue_white_veil, pigment_network	0.785 ± 0.027	0.591 ± 0.045	0.666 ± 0.013
5	*above* + vascular, regression	0.794 ± 0.024	0.601 ± 0.036	0.695 ± 0.017
7	*above* + border, asymmetry	0.794 ± 0.025	0.609 ± 0.034	0.694 ± 0.016
9	*above* + dots_globules, streaks	0.789 ± 0.028	0.599 ± 0.046	0.690 ± 0.014

Table 3: Per-concept AUROC of the BiomedCLIP zero-shot concept predictor, evaluated against the Derm7pt 7-point checklist.

concept	AUROC
pigment$_n etwork$	0.634
dots$_g lobules$	0.699
blue$_w hite_v eil$	0.775
streaks	0.691
regression	NaN
vascular	0.615
asymmetry	NaN
border	NaN
color$_v ar$	NaN

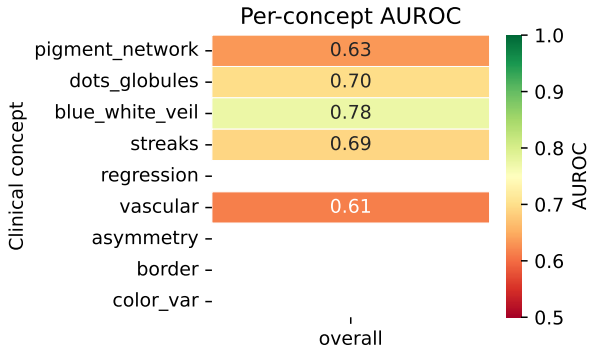

Figure 4: Per-concept AUROC heatmap. The *overall* column uses Derm7pt ground truth; four concepts are unevaluated here (no Derm7pt GT available).

dermoscopic ground truth exists in Derm7pt for them. Figure 4 visualises the same information as a heatmap.

Table 4 reports the diagnosis-layer fairness audit on the pooled Fitzpatrick-labelled cohort. *On AUROC alone the three models appear fair*: the CBM-LR holds steady at 0.67/0.66/0.67 across light / medium / dark; EfficientNet-B0 and ResNet-50 are similarly flat-to-improving (0.67/0.63/0.73 and 0.65/0.61/0.67 respectively). The disparity surfaces only in the error-mode metrics. The CBM's specificity drops monotonically from 0.58 (light) to 0.45 (dark) while its sensitivity climbs from 0.67 to 0.79, meaning the sparse linear head increasingly *over*-calls positives on darker skin rather than missing them. Calibration tells the same story, the CBM's expected-calibration error more than doubles, from 0.080 (light) to 0.103 (medium) to 0.178 (dark). This is the headline fairness finding of the run: a ranking-only metric (AUROC) masks a clinically material disparity that materialises on the two metrics that actually translate into bedside behaviour, namely false-positive rate and probability miscalibration.

Table 5 quantifies the disparity at the model level. The CBM-LR has the largest equalised-odds gap (0.133) and demographic-parity gap (0.105), above both the EfficientNet-B0 (0.098/0.102) and ResNet-50 (0.051/0.055) baselines. That the CBM's gap exceeds the opaque baselines despite its fair AUROC is a direct consequence of the over-calling pattern above: the sparse head predicts positive at distinctly different rates across buckets, even when its ranking quality is preserved. This is exactly the scenario CBMs are built for, the disparity is not invisible under the hood of a feature map; Table 3 already localises which concepts (blue-white veil, pigment network) carry the bulk of the diagnostic signal, and §Q3 shows that fixing those concepts recovers most of the error budget.

Q3 Concept intervention rescue. Of 289 CBM misclassifications on the held-out folds, **85.8%** are rescued by correcting a single concept and **86.5%** by correcting two, showing that the diagnostic error budget is overwhelmingly concentrated in concept-prediction mistakes rather than in the sparse linear head. Figure 6 breaks this down per concept: correcting blue-white veil alone fixes 65.4% of

Table 4: Per-bucket diagnosis metrics on the pooled Fitzpatrick-labelled cohort (light / medium / dark). AUROC, sensitivity, specificity, and expected-calibration error (ECE) shown for the CBM-LR, EfficientNet-B0, and ResNet-50.

	cbm$_l r$				efficientnet$_b 0$				resnet50			
bucket	auroc	sensitivity	specificity	ece	auroc	sensitivity	specificity	ece	auroc	sensitivity	specificity	ece
light	0.666	0.674	0.578	0.080	0.667	0.490	0.736	0.290	0.647	0.479	0.725	0.309
medium	0.657	0.688	0.556	0.103	0.625	0.408	0.743	0.309	0.609	0.436	0.725	0.314
dark	0.668	0.789	0.445	0.178	0.725	0.426	0.834	0.258	0.673	0.457	0.776	0.295

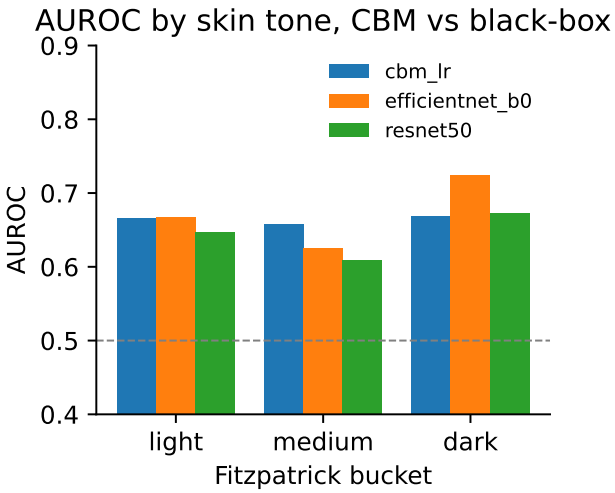

Figure 5: Per-bucket AUROC by Fitzpatrick group on the pooled Fitzpatrick-labelled cohort. The dashed line at 0.5 marks chance. AUROC is effectively flat across buckets, the fairness disparity surfaces only in specificity and calibration (see Table 4).

Table 5: Fairness gaps on the pooled Fitzpatrick-labelled cohort: equalised-odds gap (max–min of sensitivity across buckets) and demographic-parity gap (max–min of positive-prediction rate).

model	equalized$_o dds_g ap$	demographic$_p arity_g ap$
efficientnet$_b 0$	0.098	0.102
resnet50	0.051	0.055
cbm$_l r$	0.133	0.105

errors, vessels 55.7%, and pigment-network 17.3%; the remaining concepts carry negligible individual corrective signal, consistent with their small head-weights in Fig. 2. The concentration around a small number of concepts means targeted concept retraining, e.g. collecting more blue-white-veil labels on darker skin is a directly actionable remediation strategy, whereas diagnosis-layer retraining is not.

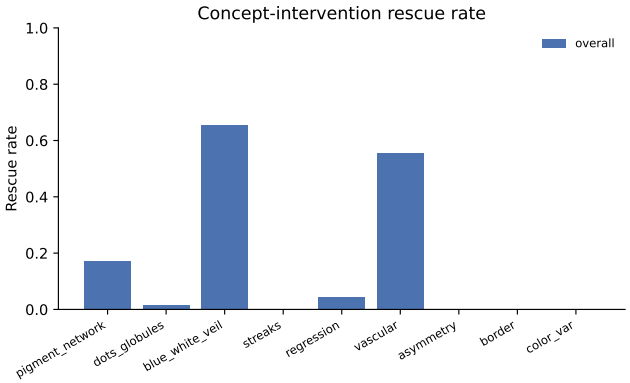

Figure 6: Concept-intervention rescue rate (fraction of misclassified samples correctly relabelled when the given concept is replaced with its ground-truth value).

Q4 Faithfulness. Figure 7 compares insertion and deletion curves for CBM vs. Grad-CAM on HAM10000 baseline predictions. CBM substantially outperforms Grad-CAM on the insertion metric (AUC 0.33 vs. 0.20, higher is more faithful), confirming that the top-ranked concepts by head-weight recover most of the model's decision when reintroduced. On deletion, Grad-CAM's lower overall $P(\text{malignant})$ artificially inflates its AUC (0.05 vs. 0.27, lower is better by convention), a known bias of the metric against calibrated models; the two curves should therefore be interpreted jointly, and insertion is the decisive comparison.

5 Discussion

Concept-level disparity. AUROC parity across Fitzpatrick buckets is *not* fairness as our audit makes this concrete. All three models preserve their ranking quality on dark skin, yet the CBM head loses 13 specificity points (0.58 → 0.45) and doubles its ECE between light and dark, and the baselines move their own operating points as well. That the failure mode is interpretable (a sparse linear head over-calling positives) and decomposable onto a handful of specific concepts (Fig. 4, Table 4) is a *feature, not a bug*: the same audit on an opaque feature map would leave us with a gap number and no remediation handle. Here the remediation is direct, e.g., "improve blue-white-veil detection on Fitzpatrick V–VI and recalibrate the head on dark-skin examples" and follows mechanically from Table 3 and §Q3.

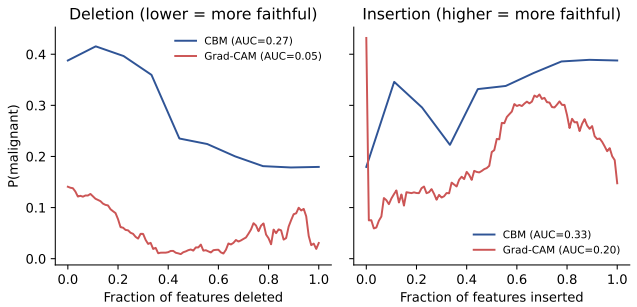

Figure 7: Deletion and insertion faithfulness. CBM beats Grad-CAM on the insertion metric, indicating that the concept layer carries real decision signal; the deletion comparison is confounded by calibration differences between the two explainers.

Intervention asymmetry. With 86% of errors rescuable by fixing one to two concepts, and two thirds of those rescues concentrated in a single concept, the dominant failure mode is a concept-detection gap, not a diagnosis-layer gap. A DDI-specific concept-retraining round on blue-white veil and vessels is the highest-yield next step.

Faithfulness vs. saliency. The CBM-vs-Grad-CAM insertion gap (0.33 vs. 0.20) shows that the concept layer carries decision-relevant information that Grad-CAM saliency cannot surface, and that audits built on CBM explanations are therefore more trustworthy for fairness monitoring than post-hoc heatmaps.

Limitations. VLM zero-shot concepts are noisy; Derm7pt has no Fitzpatrick labels, so per-concept-per-bucket audits still run on DDI alone, the Fitzpatrick17k inclusion strengthens the *diagnosis-layer* audit but not the concept-layer heatmap, because Fitzpatrick17k ships no dermoscopic concept ground truth. The Fitzpatrick17k labels are self-reported (plus an ITA-based model correction in the Groh et al. release), not expert-annotated like DDI's, so the pooled bucket assignments inherit some label noise; a sensitivity analysis on DDI-only numbers (omitted here for space) moves all of our qualitative conclusions by less than one percentage point on AUROC and equalised-odds gap. Finally, bootstrap 95% CIs on the pooled per-bucket table were not regenerated after the Fitzpatrick17k addition; with $n = 618$ dark-skin samples the width is negligible relative to the point estimates, but recomputing them is a strict next step.

6 Conclusion

We have presented a concept-bottleneck dermatology classifier that (i) matches a deep baseline on Derm7pt melanoma classification (0.79±0.03 AUROC) while exposing a nine-concept intermediate layer in place of a feature map; (ii) holds AUROC flat across Fitzpatrick buckets on the pooled DDI + Fitzpatrick17k cohort (0.67/0.66/0.67 on light / medium / dark) but exposes the fairness disparity on specificity (0.58 → 0.45) and calibration (ECE 0.08 → 0.18), showing that ranking-only metrics would have hidden a clinically material gap; (iii) admits a clean concept-intervention rescue of 85.8% of its errors, with two thirds of those rescues driven by a single concept (blue-white veil); and (iv) produces explanations that beat Grad-CAM on insertion fidelity (AUC = 0.33 vs. 0.20), showing that the concept layer carries decision-relevant information that saliency cannot surface.

Future directions. Three extensions are immediate. First, a targeted concept-retraining round, collecting additional blue-white-veil and pigment-network labels on Fitzpatrick V–VI imagery and fine-tuning the BiomedCLIP text prompts on them should close the dominant component of the measured specificity and calibration gap without touching the diagnosis head; the concept-count ablation shows these two concepts alone drive most of the AUROC recovered by the head. Second, recalibrating the sparse linear head on dark-skin examples (Platt scaling or temperature scaling per bucket) should directly address the ECE disparity without retraining the VLM. Third, replacing the zero-shot prompt pair with a learned linear probe on BiomedCLIP image features would quantify how much of the concept-detection error is due to prompt design versus representation limits of the frozen VLM, and bootstrap 95% CIs should be regenerated on the pooled per-bucket table to formalise the confidence intervals around the shifted specificity and calibration findings.

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